# Data Visualization Task Report

## Findings, Methodology, and Tools

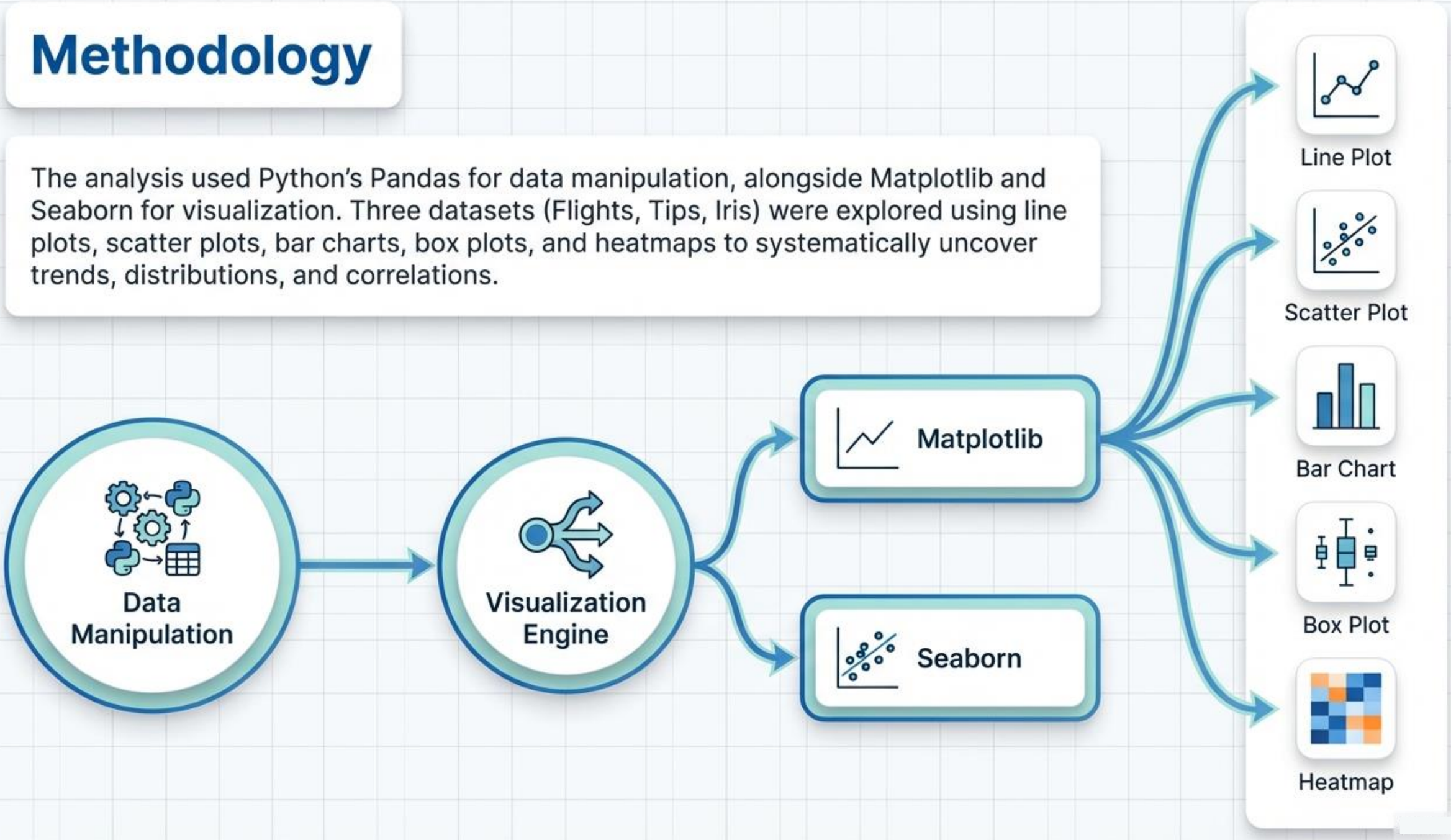
Submitted to: IEEE SSCS | AI Sub-Team

Prepared for: Omar Ashraf

Prepared by: Salah Khafaga

# Methodology

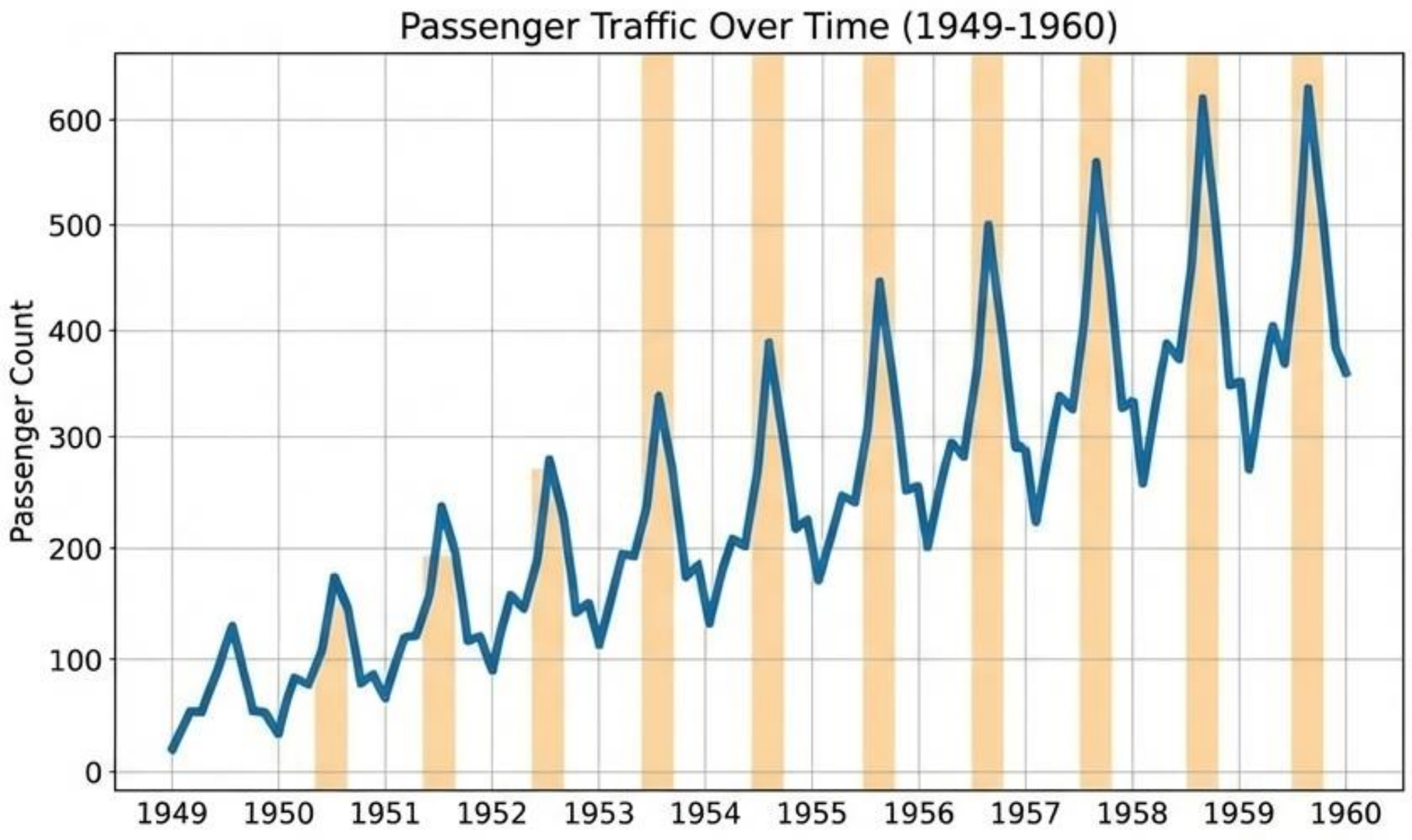
The analysis used Python's Pandas for data manipulation, alongside Matplotlib and Seaborn for visualization. Three datasets (Flights, Tips, Iris) were explored using line plots, scatter plots, bar charts, box plots, and heatmaps to systematically uncover trends, distributions, and correlations.



# Flights Dataset Findings

**Upward Trend:** A clear, consistent upward trend in airline passengers from 1949 to 1960.

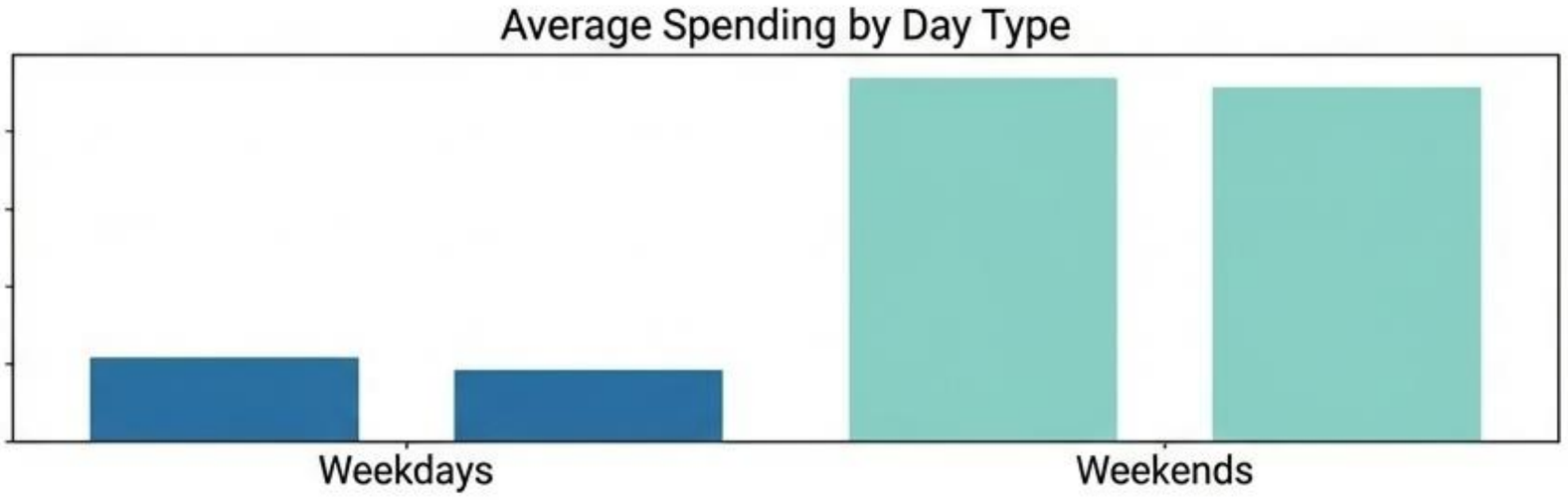
**Seasonality:** Passenger traffic heavily peaks during the summer months (July and August).

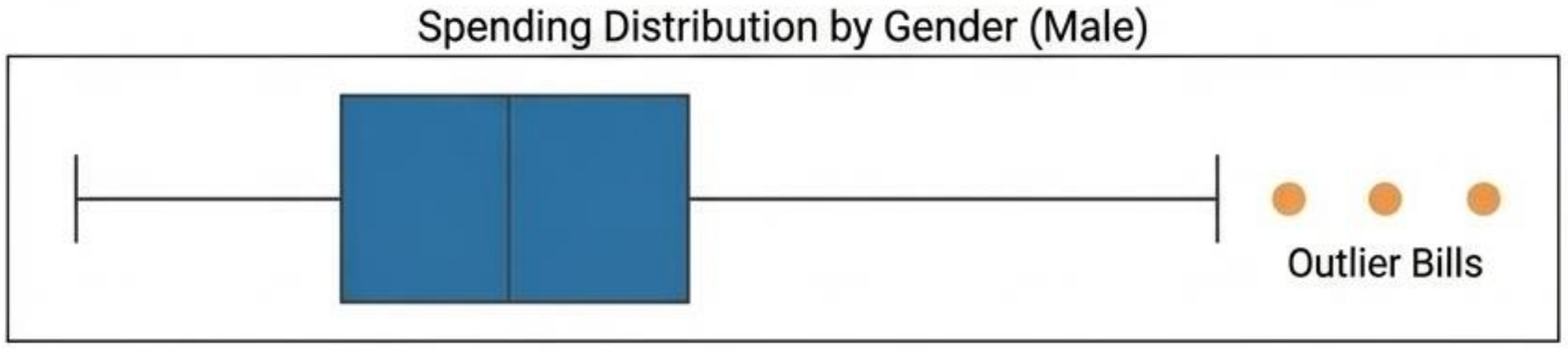


# Tips Dataset Findings

**Weekend Peaks:** Customer spending is significantly higher on weekends.

**Outliers:** Male customers have a wider distribution of spending and generate the highest outlier bills.

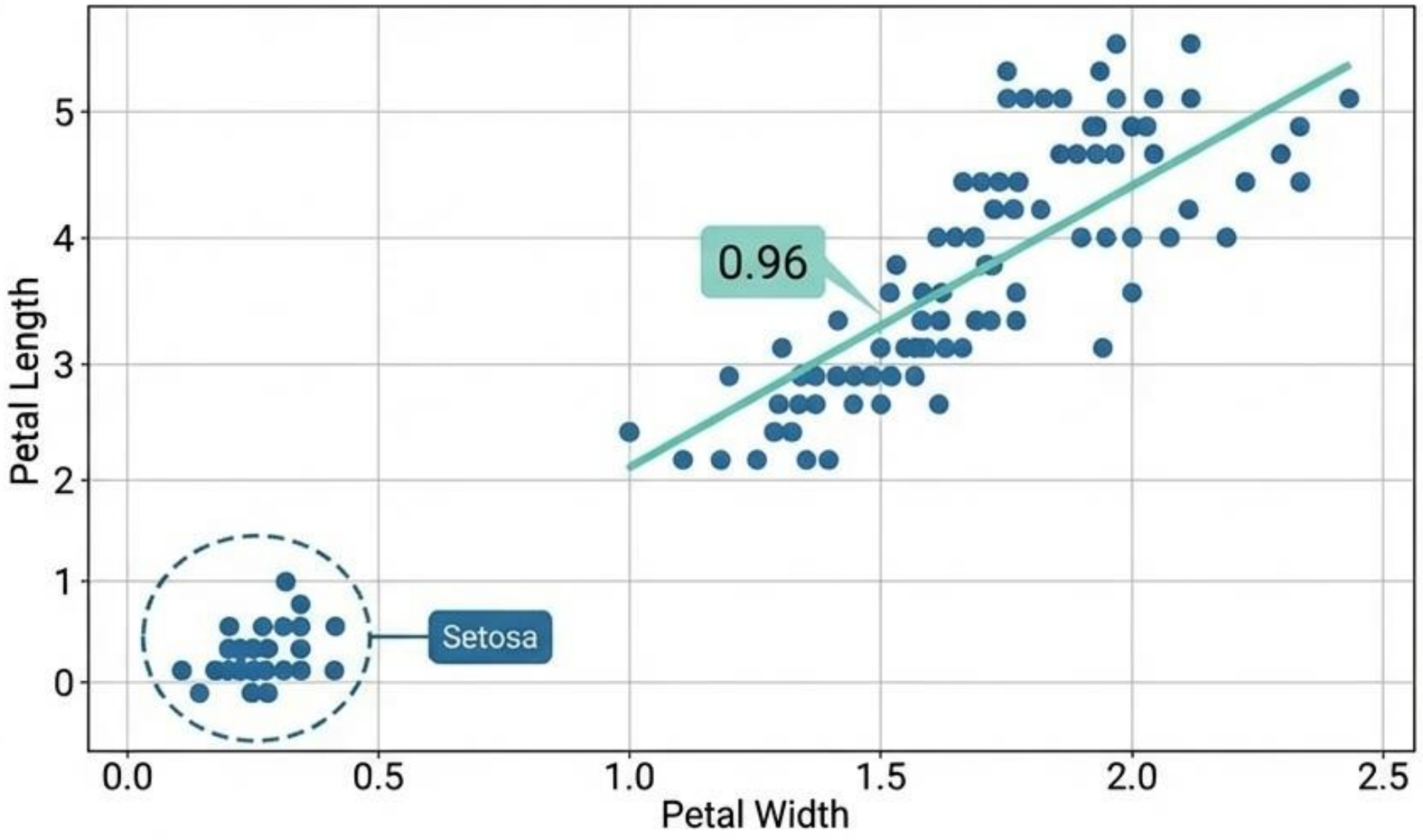


# Iris Dataset Findings

**Distinct Species:** The "Setosa" species is physically highly distinct from the others, marked by a much smaller petal length.

**High Correlation:** A near-perfect positive correlation (0.96) exists between petal length and petal width.



# Conclusions

Airplane

Restaurant Receipt

Flower

**Storytelling with Data**

Effective data visualization is the key to storytelling with data. By choosing the right chart types and libraries, raw numerical data was successfully translated into clear, actionable insights regarding historical trends, trends, human spending behaviors, and biological classifications.

# Matplotlib vs. Seaborn

## Matplotlib

- A foundational, low-level library.
- Highly customizable for controlling every pixel.
- Best for unconventional charts or precise layouts using its OOP API.

## Seaborn

- A high-level library built on Matplotlib.
- Designed for statistical visualization.
- Best for quick Exploratory Data Analysis (EDA) and attractive charts with minimal code.

# Plotly & Interactivity

**Plotly** is an advanced, open-source graphing library specializing in interactive visualizations. Unlike Matplotlib and Seaborn (static images), Plotly renders graphs using HTML/JavaScript.

- **Users can zoom, pan, and hover over data points.** It is the industry standard for building interactive data dashboards and web applications.